

Objective-Aware Projection Policies for Inverse PDE Posterior Sampling

A Two-Axis Study of Schedule and Order under Partial Observations

Hansheng Zhu
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Abstract

This report studies inverse PDE posterior sampling under sparse observations, a setting common in data assimilation and scientific sensing where uncertainty, physical validity, and runtime all matter. DiffusionPDE introduced generative PDE solving under partial observation, and Physics-Constrained Flow Matching (PCFM) introduced projection-based constraint correction during sampling. Building on these ideas, I examine which projection schedule (when to project) and projection order (how strongly to project) best trade off inverse reconstruction quality, physical consistency, runtime, and trajectory stability. The nonlinear elliptic benchmark

$$-\Delta v + \kappa v^3 = u, \quad \kappa = 50, \quad N_x = 63,$$

is used with periodic boundary conditions, where full u is recovered from sparse v observations. Relative to baseline, the best tuned setting improves u_error by **21.75%** ($1.9232 \rightarrow 1.5050$) with a clear quality-runtime tradeoff. I then extend the same protocol across additional PDE families, and the same principle appears: projection policy should be objective-aware, not globally fixed. More broadly, the results show that schedule/order is a practical control knob for flow-matching and other constrained generative samplers, because correction effort can be allocated across the trajectory to match application priorities.

1 Concepts Used

MEAM 4600 concepts used include PDE modeling, inverse formulation under partial observations, residual-based numerical correction, and objective-wise tradeoff analysis. I use these concepts to structure model design, inverse inference, and objective-wise evaluation.

Core formulas:

- PDE model: $-\Delta v + 50v^3 = u$ with periodic BC.
- Inverse target: recover full u from partial v observations.
- Residual: $r(u, v) = -\Delta v + 50v^3 - u$.

- Sampling + guidance:

$$x_{k+1} = x_k + \Delta t f_\theta(x_k, t_k) - g \nabla_x \mathcal{L}_{\text{obs}}(x_k).$$

- Projection:

$$x_{k+1} \leftarrow x_{k+1} - P_q^{-1}(x_{k+1}) r(x_{k+1}),$$

with first-order and Gauss-Newton variants.

2 Problem Framing

In inverse PDE settings, sparse measurements make recovery inherently ambiguous: many full fields can match the same observed subset. Posterior sampling is therefore used instead of a single deterministic predictor, so uncertainty can be represented and then guided toward physically plausible solutions.

Given that context, the research question is: *which projection schedule and projection order are best for each target objective?* The focus is on mechanism and tradeoffs, so the goal is principled comparison rather than state-of-the-art claims.

3 Method Summary

The benchmark studied is

$$-\Delta v + 50v^3 = u, \quad N_x = 63, \quad \text{periodic BC.}$$

Only 10% of v is observed, and full u is recovered. This observation mask is kept fixed across comparisons so schedule/order remains the main changing factor; I keep this fixed-mask protocol to isolate schedule/order effects cleanly.

The joint state is represented as $x = [u; v] \in \mathbb{R}^{2N_x}$, with residual

$$r(x) = -\Delta v + 50v^3 - u.$$

At each sampling step, the state first moves along the learned flow with observation guidance:

$$x_{k+1} = x_k + \Delta t f_\theta(x_k, t_k) - g \nabla_x \mathcal{L}_{\text{obs}}(x_k),$$

then optional projection is applied:

$$x_{k+1} \leftarrow x_{k+1} - P_q^{-1}(x_{k+1}) r(x_{k+1}),$$

where schedule S controls *when* projection is applied, and order $q \in \{\text{first_order}, \text{gauss_newton}\}$ controls *how* each correction is computed. These two axes are separated because timing and correction strength affect different failure modes (instability, over-correction, or weak physics enforcement).

The method is intentionally simplified: one prior family, one inverse formulation, and controlled schedule/order/guidance sweeps. This scope keeps conclusions interpretable and reproducible.

Baseline definition. Baseline means the default flow-matching model trained for 80 epochs at default width/depth, evaluated on the same projection grid. This keeps the comparison fair: gains come from tuning and projection policy, not from framework changes.

4 Metrics and Protocol

To make the evaluation logic explicit before presenting numbers, metric roles are first mapped to their exact function in the inverse task.

Table 1: Metric roles in this inverse setting.

Metric	Interpretation
<code>u_error</code>	Primary inverse target quality
<code>v_error</code>	Full-state reconstruction, incl. unobserved regions
<code>obs_error</code>	Fit on observed entries only
<code>posterior_quality</code>	Composite ranking score
<code>physical_consistency</code>	Pre-cleanup PDE consistency
<code>runtime / trajectory_stability</code> <code>mmse, smse, fpd</code>	Efficiency / robustness Distribution-level comparison metrics
<code>ce_ic, ce_bc, ce_cl</code>	Constraint-specific errors (N/A when inapplicable)

Evaluation uses $128 \text{ samples} \times 3 \text{ observation seeds}$ in the nonlinear main study. Observation-guidance strength is swept up to 16, then baseline is compared with a stronger checkpoint (`e200_big`: 200 epochs, larger model width/depth) under identical settings. This isolates whether gains come from training, guidance strength, or projection policy.

5 Results: Nonlinear Elliptic

5.1 Objective-wise winners

The table below gives the best projection policy per objective under one fixed evaluation setup.

These winners show why one global policy is insufficient: different objectives prefer different projection behavior. In plain words: the stronger checkpoint is a larger model

Table 2: Best schedule/order by objective.

Objective	Winning run	Best policy	Value
<code>posterior_quality</code>	larger 200-epoch checkpoint + strong guidance	project every 5 steps / Gauss-Newton correction	3.1364
<code>physical_consistency</code>	larger 200-epoch checkpoint + strong guidance	project every step / Gauss-Newton correction	0.000359
<code>runtime</code>	baseline (default 80-epoch checkpoint)	no projection	0.6086 s
<code>trajectory_stability</code>	baseline (default 80-epoch checkpoint)	no projection	0.0

trained for 200 epochs; strong guidance means a high weight on matching observations; “project every step” and “project every 5 steps” define correction frequency; first-order and Gauss-Newton are two correction rules; “no projection” means no correction pass.

Key result. The best inverse-reconstruction setup uses the larger 200-epoch model, strong observation guidance (strength 16), and first-order projection at every sampling step. It reaches $u_error = 1.5050$ and $v_error = 0.9120$, improving baseline by **21.75%** and **29.54%**.

5.2 Best inverse row vs baseline

To evaluate inverse recovery directly, the best- u row from baseline is compared against the best- u row after tuning.

Table 3: Best inverse row comparison (128×3).

Setting	<code>u_err</code>	<code>v_err</code>	<code>obs_err</code>	<code>post_q</code>	<code>runtime</code>
Baseline best row	1.9232	1.2944	1.1504	4.3679	0.7357 s
Best tuned (strong guidance, projection every step, first-order correction)	1.5050	0.9120	0.8126	3.2295	1.3741 s
Relative change	-21.75%	-29.54%	-29.36%	-26.06%	+86.77%

Best inverse rows are compared directly because u_error is the primary metric for the inverse target. This row-to-row comparison shows substantial reconstruction gains and also quantifies the runtime cost of stronger correction.

Key nonlinear conclusion. If the goal is best inverse reconstruction, project at every step with first-order correction. If the goal is best physics consistency, use Gauss-Newton-heavy correction. If the goal is speed or maximum stability, skip projection.

6 Results: Multi-Family Extension

To test whether the nonlinear finding is specific to one PDE, the same pipeline is applied to linear elliptic, heat, reaction-diffusion, Burgers (IC/BC), and a 1D Navier-Stokes surrogate. The same evaluation logic is kept so cross-family conclusions remain comparable.

The following table reports how much inverse-target error drops as guidance strength increases from weak to strong.

Table 4: Best-row u_error gain from $g = 0.25$ to $g = 16$.

Family	u_error reduction
Linear elliptic	-18.73%
Heat equation	-25.20%
Reaction-diffusion	-29.84%
Burgers IC	-26.12%
Burgers BC	-9.03%
Navier-Stokes 1D surrogate	-24.11%

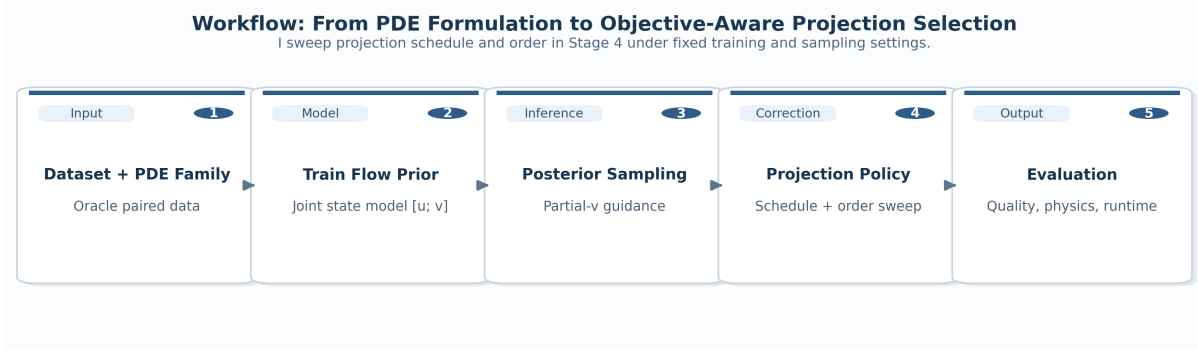


Figure 1: Workflow from PDE-family data to objective-wise schedule/order ranking. Projection is inserted inside the sampling trajectory and swept systematically.

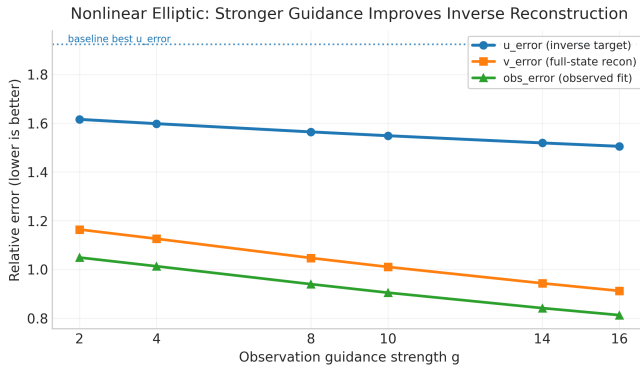


Figure 2: Stronger guidance consistently improves reconstruction metrics.

Across families, stronger observation guidance consistently improves inverse recovery, while exact schedule/order winners remain objective-specific. This supports the same design principle from the core benchmark.

7 Reflection and Reproducibility

What worked. Material nonlinear improvement was achieved, and complete objective-wise ranking tables were produced.

What did not always work. Larger/longer training was not universally better across families, so capacity alone should not be assumed to solve inverse quality.

What I learned. Projection policy should be selected by objective, and low residual after cleanup should not be treated as a proxy for good inverse-target recovery.

Attribution and responsibility. Existing repository components (especially `src/chonkdiff`) were used as a numerical backend, and the posterior-projection study was implemented/extended in `src/posterior_projection`. LLM assistance was used for coding/report drafting, and metric values, commands, and final conclusions were manually verified before submission.

8 Limitations and Next Steps

The method was intentionally simplified, which improved interpretability but limits direct comparability to larger

benchmark suites. In particular, full-scale reproduction of all external baselines and exhaustive architecture searches were not attempted. A natural next step is to reuse this same evaluation protocol in broader cross-method studies (using `mmse/smse/fpd` and family-specific constraint errors) while preserving the objective-wise projection analysis developed here.

9 Conclusion

Projection schedule/order is a first-class algorithmic choice in partial-observation inverse PDE sampling. Results show that a larger 200-epoch model, strong observation guidance (strength 16), and first-order projection at every sampling step gives the strongest inverse recovery on the proposal benchmark, while other policies win other objectives. Therefore, the practical rule is clear: choose projection design to match the objective of interest, rather than applying one fixed policy everywhere.

References

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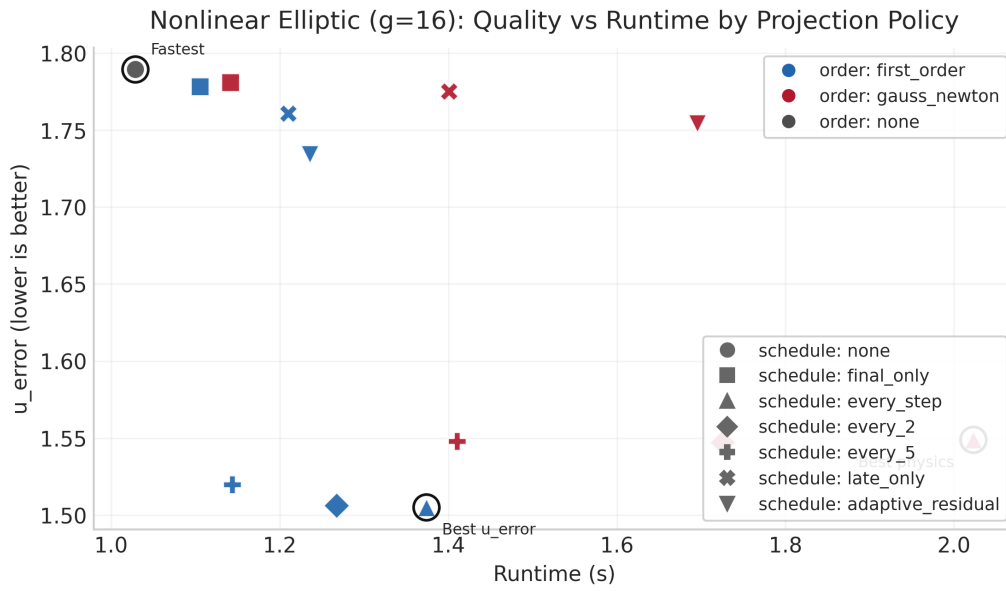


Figure 3: Quality-runtime frontier at guidance strength 16. Color shows correction rule, marker shows projection schedule, and rings mark best inverse quality, fastest runtime, and best physics consistency.

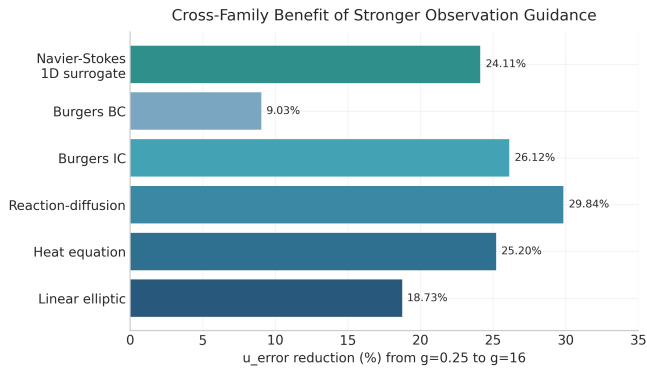


Figure 4: Cross-family trend of stronger-guidance benefit.